

File Edit Selection View Go Run Terminal Help python_test

EXPLORER

PYTHON TEST

- exercise2.panorama.py
- exercise2.py
- exercise2.target.py
- exercise3.animation.py
- exercise3.chessboard.py
- exercise3.day3.py
- exercise3.py
- exercise3.toolmask.py
- exercise4.annotations....
- exercise4.day3.py
- exercise4.smart.py
- exercise4.splash.py
- exercise5_license_plat...
- exercise5.alignment.py
- exercise5.clock.py
- exercise5.mask.py
- face_detector.py
- family photo.jpg
- final_collage.png
- flipping.py
- getting_and_setting.py
- gradient.png
- grayscale_histogram.p...
- grayscale_histogram.py
- histogram_equalizatio...
- histogram_search_de...
- histogram_with_mask...
- house.png
- hsv_color_wheel.png
- image.enhancement.c...
- image.py
- imutils.py
- load_display_save.py

face_detector.py

```
1 #face_detector.py
2 # Face Detector class using Haar cascades
3 import cv2
4
5 class FaceDetector:
6     """ A class for detecting faces in images using Haar cascades """
7
8     def __init__(self, faceCascadePath):
9         """ Initialize the face detector with a cascade classifier
10         Parameters:
11         - faceCascadePath: path to the XML cascade file
12         """
13         # Load the cascade classifier
14         self.faceCascade = cv2.CascadeClassifier(faceCascadePath)
15
16         # Check if cascade was loaded properly
17         if self.faceCascade.empty():
18             raise ValueError(f"Could not load cascade from: {faceCascadePath}")
19         else:
20             print("✓ Face cascade loaded successfully")
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python face_detector.py
PS C:\Users\raghu\OneDrive\Desktop\python_test>

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EXPLORER

PYTHON_TEST

days_transformations...
day6_arithmetic_bitwis...
day7_mini.challenges.py
day8_color_spaces.py
day9_histograms.py
day10_smoothing_blu...
day11_thresholding.py
day13_contours.py
day14_mini_project.py
detect_faces.py
dissolve.mp4
drawing_basics.py
drawing.py
edge_preserving_com...
enhanced_image.png
equalization_comparis...
equalize.py
equalized_image.png
exercise_1_cascade_co...
exercise_1_channel_art...
exercise_1_contour_an...
exercise_1_custom_an...
exercise_1_custom_pip...
exercise_1_histogram_...
exercise_1_interactive_...
exercise_1_threshold_...
exercise_2_batch_dete...
exercise_2_bounding_...
exercise_2_color_picke...
exercise_2_compariso...
exercise_2_document_...
exercise_2_equalizatio...
exercise_2_noise radi...

OUTLINE
TIMELINE

Welcome

exercise_2_batch_detection.py X

exercise_2_batch_detection.py > ...

1 # exercise_2_batch_detection.py
2
3 import cv2
4 import os
5 import glob
6
7
8 def batch_face_detection(image_dir, cascade_path, output_dir="detected_faces"):
9 """Detect faces in all images in a directory"""
10
11 # Create output directory
12 if not os.path.exists(output_dir):
13 os.makedirs(output_dir)
14
15 # Load cascade
16 cascade = cv2.CascadeClassifier(cascade_path)
17 if cascade.empty():
18 print("Could not load cascade")
19 return
20
21 # Get all images

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_2_batch_detection.py
Found 3 images
family photo.jpg: 3 faces detected
messi.jpg: 2 faces detected
obama.png: 1 faces detected
PS C:\Users\raghu\OneDrive\Desktop\python_test>

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EXPLORER

WELCOME

exercise_3_face_roi.py

PYTHON_TEST

exercise_1_custom_pip...

exercise_1_histogram...

exercise_1_interactive...

exercise_1_threshold...

exercise_2_batch_dete...

exercise_2_bounding...

exercise_2_color_picke...

exercise_2_compariso...

exercise_2_document...

exercise_2_equalizatio...

exercise_2_noise_redu...

exercise_2_object_cou...

exercise_3d_color_hi...

exercise_3_color_isolat...

exercise_3_crop_objec...

exercise_3_custom_ker...

exercise_3_face_roi.py

exercise_3_multi_objec...

exercise_3_size_classifi...

exercise_4_advanced_s...

exercise_4_backgroun...

exercise_4_color_spac...

exercise_4_edge_prese...

exercise_4_histogram...

exercise_4_shape_clas...

exercise_5_complete_a...

exercise_5_contrast_en...

exercise_5_deblurring...

exercise_5_document...

exercise1.brighthness.c...

exercise1.day3.py

exercise1.house.py

exercise1.nu...

OUTLINE

TIMELINE

exercise_3_face_roi.py > ...

1# exercise_3_face_roi.py

2

3import cv2

4

5

6def detect_faces_with_roi(image_path, cascade_path, roi=None):

7 """Detect faces in a specific region of interest"""

8

9 image = cv2.imread(image_path)

10 if image is None:

11 return []

12

13 cascade = cv2.CascadeClassifier(cascade_path)

14 gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

15

16 # If ROI is specified, crop the image

17 if roi is not None:

18 x, y, w, h = roi

19

20 roi_image = gray[y:y+h, x:x+w]

21

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_3_face_roi.py

Found 0 faces in ROI

PS C:\Users\raghu\OneDrive\Desktop\python_test>

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EXPLORER

PYTHON_TEST

drawing_basics.pydrawing.pyedge_preserving_com...enhanced_image.pngequalization_comparis...equalize.pyequalized_image.pngexercise_1_cascade_co...exercise_1_channel_art...exercise_1_contour_an...exercise_1_custom_an...exercise_1_custom_pip...exercise_1_histogram...exercise_1_interactive...exercise_1_threshold_...exercise_2_batch_dete...exercise_2_bounding_...exercise_2_color_picke...exercise_2_compariso...exercise_2_document_...exercise_2_equalizatio...exercise_2_noise_redu...exercise_2_object_cou...exercise_3_d_color_hi...exercise_3_color_isolat...exercise_3_crop_objec...exercise_3_custom_ker...exercise_3_face_roi.pyexercise_3_multi_objec...exercise_3_size_classifi...exercise_4_advanced_s...exercise_4_backgroun...

OUTLINETIMELINE

Welcomeexercise_3_face_roi.pyexercise_4_performance_test.py

exercise_4_performance_test.py > ...

```
1 # exercise_4_performance_test.py
2
3 import cv2
4 import time
5 import numpy as np
6
7
8 def performance_test(image_path, cascade_path):
9     """Test face detection performance with different parameters"""
10
11     image = cv2.imread(image_path)
12
13     if image is None:
14         print(f"Could not load image: {image_path}")
15         return
16
17     gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
18
19     cascade = cv2.CascadeClassifier(cascade_path)
20
21     if cascade.empty():
```

PROBLEMSOUTPUTDEBUG CONSOLETERMINALPORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_4_performance_test.py

Slow, accurate: 2 faces in 2936.3ms

Balanced: 1 faces in 24.5ms

Fast, may miss faces: 1 faces in 14.6ms

Very fast: 1 faces in 9.9ms

Accurate, fewer false positives: 1 faces in 24.0ms

Most accurate, slowest: 1 faces in 43.0ms

PS C:\Users\raghu\OneDrive\Desktop\python_test>

+ ...

powershell

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